CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 6 : Python Program for Vacuum Cleaner Problem

Aim

To develop a Python program for the Vacuum Cleaner problem, where the agent visits every room and cleans it if it is dirty.

Algorithm

- Read the status of all rooms as a list (0 = clean, 1 = dirty).
- Visit the rooms one by one from the first to the last.
- If the current room is dirty (1), display that it is dirty and clean it by setting its status to 0.
- If the current room is already clean (0), display that it is clean.
- After all rooms are visited, display "All rooms cleaned".

Program

```
rooms = input("Enter status of rooms (0 clean, 1 dirty): ").split()

for i in range(len(rooms)):
if rooms[i] == "1":
        print("Room", i + 1, "is dirty")
print("Cleaning room", i + 1)
rooms[i] = "0"     else:
print("Room", i + 1, "is clean")
print("All rooms cleaned")
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============== RESTART: C:/Users/shaik/OneDrive/Documents/6.py ==============
Enter status of rooms (0 clean, 1 dirty): 1 0 1 1 0
Room 1 is dirty
Cleaning room 1
Room 2 is clean
Room 3 is dirty
Cleaning room 3
Room 4 is dirty
Cleaning room 4
Room 5 is clean
All rooms cleaned
>>>
```

RESULT

Thus, the Vacuum Cleaner problem was solved successfully using Python.